

Functional Specification

Status legend: **Implemented** · **Partial** · **Stub** / **Planned**

1. Authentication & session

Capability	Status	Notes
Staff login (username/password)	Implemented	JWT returned
Current user profile (/auth/me)	Implemented	
Change password	Implemented	Min 8 chars; confirm match
Remember-me (local vs session storage)	Implemented	Frontend
Failed-attempt logout	Implemented	LoginAttemptService
Role-based route guards	Implemented	Angular authGuard

2. Dashboard

Capability	Status	Notes
Summary counts (customers, accounts, employees)	Implemented	
Today's transaction count	Partial	Hardcoded 0 until Transaction module exists

3. Customers

Capability	Status	Notes
Search / paginate customers	Implemented	
View customer detail	Implemented	
Create customer	Implemented	Optional first account on create
Update customer	Implemented	
Soft/hard delete	Implemented	ADMIN only for DELETE
Open additional accounts from detail	Implemented	Current (policy-aware opening deposit); Loan button present

4. Accounts

Capability	Status	Notes
Open account	Implemented	Policy-validated opening deposit
List / search accounts	Implemented	/app/accounts
List accounts by customer	Implemented	
Get account by id	Implemented	GET /accounts/{id}
Update account status	Implemented	API present
Deposit (post-open)	Implemented	Credits balances + writes CREDIT ledger row; Kafka email to customer
Withdraw	Implemented	DEBIT + Kafka email
Transfer	Implemented	DEBIT+CREDIT; Kafka email to destination customer
Account detail + ledger UI	Implemented	Account detail + paged transactions
List pagination	Implemented	First/Last + Go to page (list-pagination)
Account policies (create + get active)	Implemented	Seeded INR policies; no LOAN seed
Opening-deposit dialog (Current)	Implemented	Uses GET /account-policies
Kafka customer email	Implemented	See MODULES/Notification.md

5. Employees (users)

Capability	Status	Notes
List / create / update employees	Implemented	ADMIN
Role assignment	Implemented	Seeded roles include unused TELLER / AUDITOR / CUSTOMER

6. Branch

Capability	Status	Notes
Branch master data	Stub	Only branchCode string (often 0001)

7. Loans

Capability	Status	Notes
Loan product / EMI / collateral	Stub	
<code>AccountType.LOAN</code> account opening	Partial	Enum + UI affordance; needs policy; not a loan domain

8. Transactions & reports

Capability	Status	Notes
Transaction ledger entity/API	Partial	Write on deposit / withdraw / transfer / opening; list + account- detail UI; dashboard count still stub
Reports	Stub	Empty <code>report</code> package; sidebar placeholder
Beneficiaries	Stub	Empty <code>beneficiary</code> package

9. Notifications

Capability	Status	Notes
Kafka bank-action events	Implemented	Topic <code>bankone.notifications</code>
Customer email on open / deposit / withdraw / transfer	Implemented	Recipient = <code>customers.email</code>
Local Mailpit / Render SendGrid HTTPS	Implemented	See <code>MODULES/Notification.md</code>

10. Audit

Capability	Status	Notes
JPA created/updated by fields	Partial	<code>AuditableEntity</code> on some entities
Audit event API / UI	Stub	Empty <code>audit</code> package

Non-functional (current)

Area	Behavior
AuthN	JWT HMAC, 1h expiry (<code>jwt.expiration</code>)
AuthZ	Role matchers in <code>SecurityConfig</code>
CORS	localhost, bankone.local, 192.168.0.4
Logging	Hibernate SQL DEBUG (dev-oriented)
Deploy	Open Liberty WAR primary path; Render Docker + Aiven + SendGrid
Messaging	Kafka (optional via <code>APP_KAFKA_ENABLED</code>)

Deferred hardening items are tracked in `.cursor/rules/deferred-hardening.mdc` .